torch.linalg.cholesky#

https://pytorch.org/docs/stable/generated/torch.linalg.cholesky.html

Description:

Computes the Cholesky decomposition of a complex Hermitian or real symmetric positive-definite matrix. Letting K\mathbb{K}K be R\mathbb{R}R or C\mathbb{C}C, the Cholesky decomposition of a complex Hermitian or real symmetric positive-definite matrix A∈Kn×nA \in \mathbb{K}^{n \times n}A∈Kn×n is defined as where LLL is a lower triangular matrix with real positive diagonal (even in the complex case) and LHL^{\text{H}}LH is the conjugate transpose when LLL is complex, and the transpose when LLL is real-valued. Supports input of float, double, cfloat and cdouble dtypes. Also supports batches of matrices, and if A is a batch of matrices then the output has the same batch dimensions.

Function Signature

```
torch.linalg.cholesky(A, *, upper=False, out=None) → Tensor#

Parameters

Keyword Arguments

Raises
```

Parameters

Keyword

upper (bool, optional) – whether to return an upper triangular matrix. The tensor returned with upper=True is the conjugate transpose of the tensor returned with upper=False. out (Tensor, optional) – output tensor. Ignored if None. Default: None.

Code Examples

```
>>> A = torch.randn(2, 2, dtype=torch.complex128)
>>> A = A @ A.T.conj() + torch.eye(2) # creates a Hermitian
positive-definite matrix
>>> A
tensor([[2.5266+0.0000j, 1.9586-2.0626j],
        [1.9586+2.0626j, 9.4160+0.0000j]],
dtype=torch.complex128)
>>> L = torch.linalg.cholesky(A)
>>> L
tensor([[1.5895+0.0000j, 0.0000+0.0000j],
        [1.2322+1.2976j, 2.4928+0.0000j]],
dtype=torch.complex128)
>>> torch.dist(L @ L.T.conj(), A)
tensor(4.4692e-16, dtype=torch.float64)

>>> A = torch.randn(3, 2, 2, dtype=torch.float64)
>>> A = A @ A.mT + torch.eye(2)  # batch of symmetric positive-
definite matrices
>>> L = torch.linalg.cholesky(A)
>>> torch.dist(L @ L.mT, A)
tensor(5.8747e-16, dtype=torch.float64)
```

Notes & Warnings

NOTE Note When inputs are on a CUDA device, this function synchronizes that device with the CPU. For a version of this function that does not synchronize, see torch.linalg.cholesky_ex().

NOTE See also torch.linalg.cholesky_ex() for a version of this operation that skips the (slow) error checking by default and instead returns the debug information. This makes it a faster way to check if a matrix is positive-definite. torch.linalg.eigh() for a different decomposition of a Hermitian matrix. The eigenvalue decomposition gives more information about the matrix but it slower to compute than the Cholesky decomposition.

Detailed Documentation

Documentation

Computes the Cholesky decomposition of a complex Hermitian or real symmetric positive-definite matrix.

Letting K\mathbb{K}K be R\mathbb{R}R or C\mathbb{C}C, the Cholesky decomposition of a complex Hermitian or real symmetric positive-definite matrix A∈Kn×nA \in \mathbb{K}^{n \times n}A∈Kn×n is defined as

where LLL is a lower triangular matrix with real positive diagonal (even in the complex case) and LHL^{\text{H}}LH is the conjugate transpose when LLL is complex, and the transpose when LLL is real-valued.

Supports input of float, double, cfloat and cdouble dtypes. Also supports batches of matrices, and if A is a batch of matrices then the output has the same batch dimensions.

When inputs are on a CUDA device, this function synchronizes that device with the CPU. For a version of this function that does not synchronize, see torch.linalg.cholesky_ex().

torch.linalg.cholesky_ex() for a version of this operation that skips the (slow) error checking by default and instead returns the debug information. This makes it a faster way to check if a matrix is positive-definite.

torch.linalg.eigh() for a different decomposition of a Hermitian matrix. The eigenvalue decomposition gives more information about the matrix but it slower to compute than the Cholesky decomposition.

A (Tensor) – tensor of shape (*, n, n) where * is zero or more batch dimensions consisting of symmetric or Hermitian positive-definite matrices.

upper (bool, optional) – whether to return an upper triangular matrix. The tensor returned with upper=True is the conjugate transpose of the tensor returned with upper=False.

out (Tensor, optional) – output tensor. Ignored if None. Default: None.

RuntimeError – if the A matrix or any matrix in a batched A is not Hermitian (resp. symmetric) positive-definite. If A is a batch of matrices, the error message will include the batch index of the first matrix that fails to meet this condition.